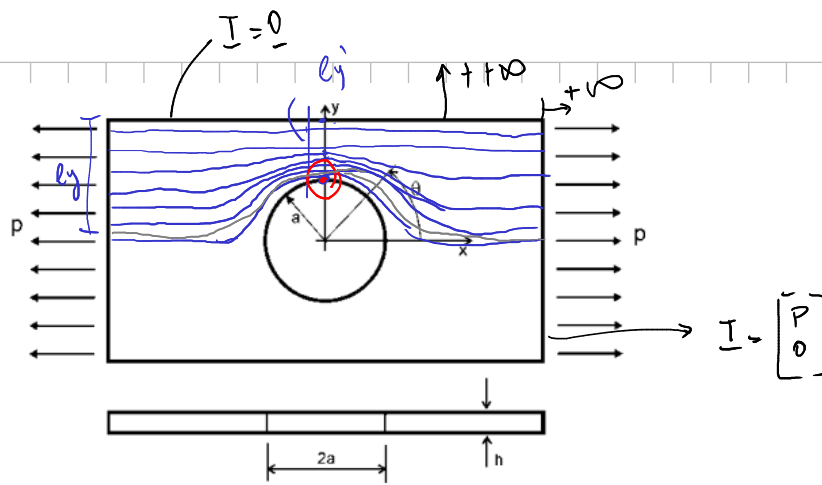
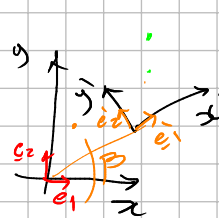


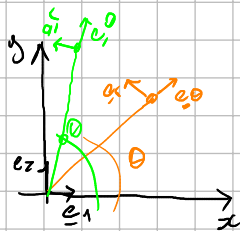
$$\sigma_r = \frac{p}{2} \left(1 - \frac{a^2}{r^2} \right) \left[1 + \left(1 - 3 \frac{a^2}{r^2} \right) \cos 2\theta \right]$$

$$\sigma_\theta = \frac{p}{2} \left[1 + \frac{a^2}{r^2} - \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right]$$

$$\tau_{r\theta} = -\frac{p}{2} \left(1 - \frac{a^2}{r^2} \right) \left(1 + 3 \frac{a^2}{r^2} \right) \sin 2\theta$$



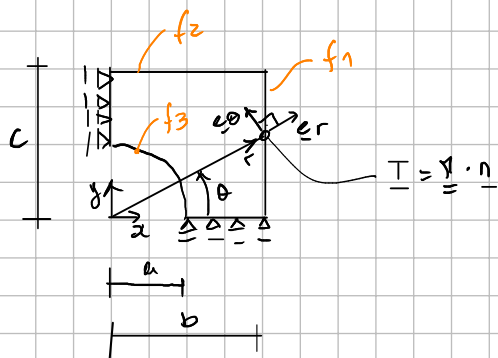
$\Rightarrow P = (p)$ es constante punto a punto



$$P(\theta) =$$

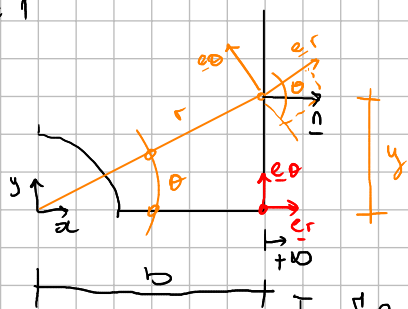
$$\begin{cases} x = r \cdot \cos \theta \\ y = r \cdot \sin \theta \\ z = z \end{cases} \Rightarrow \begin{cases} \theta = \tan^{-1} \left(\frac{y}{x} \right) \\ r^2 = x^2 + y^2 \\ z = z \end{cases}$$

2)



$$\|\underline{n}\| = 1$$

• Frontera 1



$$\underline{n} = \begin{bmatrix} n_r \\ n_\theta \end{bmatrix} = \begin{bmatrix} \cos \theta \cdot 1 \\ -\sin \theta \cdot 1 \end{bmatrix} = \begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix}$$

$$\underline{T} = \underline{\gamma} \cdot \underline{n} = \begin{bmatrix} \gamma_r & \gamma_\theta \end{bmatrix} \cdot \begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix} = \begin{bmatrix} \gamma_r \cos \theta - \gamma_\theta \sin \theta \\ \gamma_r \sin \theta + \gamma_\theta \cos \theta \end{bmatrix} = \begin{bmatrix} T_r \\ T_\theta \end{bmatrix}$$

• T_r

$$\sigma_r = \frac{p}{2} \left(1 - \frac{a^2}{r^2} \right) \left[1 + \left(1 - 3 \frac{a^2}{r^2} \right) \cos 2\theta \right] \quad r^2 = b^2 + y^2$$

$$\begin{aligned} \gamma_r(r, \theta) &= \lim_{b \rightarrow +\infty} \frac{p}{2} \left(1 - \frac{a^2}{b^2 + y^2} \right) \cdot \left[1 + \left(1 - 3 \frac{a^2}{b^2 + y^2} \right) \cdot \cos \left(2 \cdot \arctan \left(\frac{y}{b} \right) \right) \right] \\ &= \frac{p}{2} [1 + 1] = p \end{aligned}$$

$$\tau_{r\theta} = -\frac{p}{2} \left(1 - \frac{a^2}{r^2} \right) \left(1 + 3 \frac{a^2}{r^2} \right) \sin 2\theta$$

$$\gamma_\theta(r, \theta) = \lim_{b \rightarrow +\infty} -\frac{p}{2} \left(1 - \frac{a^2}{b^2 + y^2} \right) \cdot \left(1 + 3 \frac{a^2}{b^2 + y^2} \right) \cdot \sin \left(2 \cdot \arctan \left(\frac{y}{b} \right) \right) = 0$$

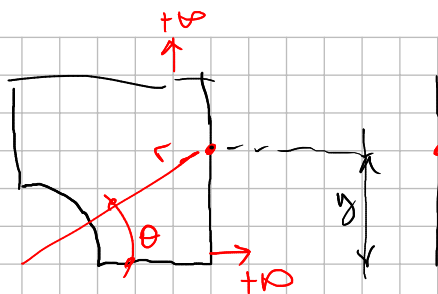
Ver continuidad de variables en θ (página 7)

$$T_r = \lim_{b \rightarrow +\infty} \gamma_r \cos \theta - \gamma_\theta \sin \theta = \lim_{b \rightarrow +\infty} p \cos \left(\arctan \left(\frac{y}{b} \right) \right) = p$$

— 0 —

$$\theta = \arctan \left(\frac{y}{x} \right) = \arctan \left(\frac{y}{b} \right) \rightarrow b \rightarrow +\infty \Rightarrow \theta = \lim_{b \rightarrow +\infty} \arctan \left(\frac{y}{b} \right) = 0$$

$$r^2 = x^2 + y^2 = b^2 + y^2 \rightarrow r = \lim_{b \rightarrow +\infty} (b^2 + y^2) = +\infty \Rightarrow r \rightarrow +\infty \Rightarrow r = b$$



$$\lim_{b \rightarrow +\infty} r_r(b, 0) = \frac{P}{2} \left(1 - \frac{a^2}{b^2} \right) \cdot \left(1 + \left(1 - 3 \frac{a^2}{b^2} \right) \cdot \overbrace{\cos(2 \cdot 0)}^1 \right) = P$$

$$\lim_{b \rightarrow +\infty} z_{re}(b, 0) = -\frac{P}{2} \left(1 - \frac{a^2}{b^2} \right) \cdot \left(1 + 3 \frac{a^2}{b^2} \right) \cdot \overbrace{\sin(2 \cdot 0)}^0 = 0$$

$$\underline{n} = \lim_{b \rightarrow +\infty} \begin{bmatrix} \overbrace{\cos \theta}^{+90} \\ \overbrace{-\sin \theta}^{+90} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

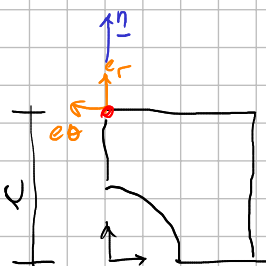
$$\underline{I} = \begin{bmatrix} r_{rr} & z_{re} \\ z_{re} & r_{\theta\theta} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} r_{rr} \\ z_{re} \end{bmatrix} = \begin{bmatrix} P \\ 0 \end{bmatrix}$$

- Frontiera f_2

$$\theta = \lim_{c \rightarrow +\infty} \arctan\left(\frac{\overbrace{c}^{+\infty}}{x}\right) = \frac{\pi}{2}$$

$$c \rightarrow +\infty \rightarrow r \rightarrow +\infty$$

$$\underline{n}_\theta = \frac{\pi}{2} = \begin{bmatrix} n_r \\ n_\theta \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$\underline{I} = \begin{bmatrix} r_r & z_{re} \\ z_{re} & r_\theta \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} r_r \\ z_{re} \end{bmatrix}$$

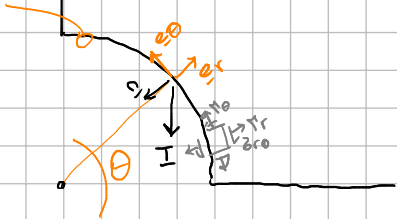
$$\lim_{c \rightarrow +\infty} r_r\left(c, \frac{\pi}{2}\right) = \frac{P}{2} \left(1 - \frac{a^2}{c^2} \right) \cdot \left[1 + \left(1 - 3 \frac{a^2}{c^2} \right) \cdot \underbrace{\cos\left(2 \cdot \frac{\pi}{2}\right)}_{-1} \right] = 0$$

$$\lim_{c \rightarrow +\infty} z_{re}\left(c, \frac{\pi}{2}\right) = -\frac{P}{2} \left(1 - \frac{a^2}{c^2} \right) \cdot \left(1 + 3 \frac{a^2}{c^2} \right) \cdot \underbrace{\sin\left(2 \cdot \frac{\pi}{2}\right)}_0 = 0$$

$$\Rightarrow \underline{I} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

circunf. $r=a \quad \theta \in [0, \pi/2]$

$$\underline{n} = \begin{bmatrix} n_r \\ n_\theta \end{bmatrix} = - \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$\underline{I} = \begin{bmatrix} T_r \\ T_\theta \end{bmatrix} = \begin{bmatrix} Y_{rr} & Z_{r\theta} \\ Z_{r\theta} & Y_{\theta\theta} \end{bmatrix} \cdot \left(- \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) = - \begin{bmatrix} Y_{rr} \\ Z_{r\theta} \end{bmatrix}$$

$\forall \theta \in [0, \pi/2]$

$$Y_{rr}(a, \theta) = \frac{p}{2} \cdot \left(1 - \frac{a^2}{a^2} \right) \cdot \left[1 + \left(1 - \frac{3a^2}{a^2} \right) \cdot \cos(2\theta) \right] = 0$$

$$Z_{r\theta}(a, \theta) = -\frac{p}{2} \cdot \left(1 - \frac{a^2}{a^2} \right) \cdot \left(1 + \frac{3a^2}{a^2} \right) \cdot \sin(2\theta) = 0$$

$$\underline{I} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \forall \theta \in [0, \pi/2]$$

b)

$$\sigma_\theta = \frac{p}{2} \left[1 + \frac{a^2}{r^2} - \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right]$$

$$\frac{\partial \sigma_\theta}{\partial r} = \frac{p}{2} \left[-\frac{2a^2}{r^3} - \left(-12 \cdot \frac{a^4}{r^5} \right) \cdot \cos(2\theta) \right] = 0$$

$$\Rightarrow r^2 = 6a^2 \cdot \cos(2\theta)$$

$$\frac{\partial \sigma_\theta}{\partial \theta} = \frac{p}{2} \left[-1 \left(1 + 3 \frac{a^4}{r^4} \right) \cdot -2 \cdot \sin(2\theta) \right] = 0$$

$$\underbrace{p \left(1 + 3 \frac{a^4}{r^4} \right)}_{\neq 0} \cdot \sin(2\theta) = 0 \Rightarrow \sin(2\theta) = 0$$

$$\Downarrow$$

$$\theta = 0 \quad \theta = \pi/2$$

± 1

$$r^2 = 6a^2 \cdot \cos\left(2 \cdot \frac{\pi}{2}\right) \Rightarrow r = \sqrt{6} a$$

$\rightarrow$ se descarta $\theta = \pi/2$ porque daría imaginario



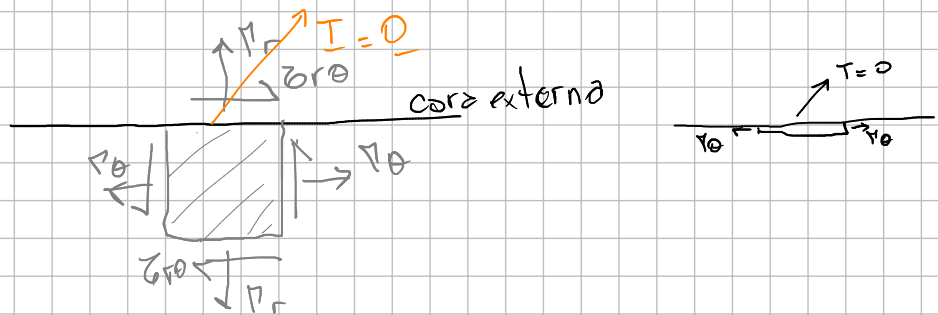
$$V_\theta(\sqrt{b}a, 0) = \frac{P}{24} (14 - 13) = \frac{P}{24}$$

• frontera f_1

$$\lim_{b \rightarrow +\infty} V_\theta(b, 0) = \frac{P}{2} \left[1 + \frac{a^2}{b^2} - \left(1 + 3 \frac{a^4}{b^4} \right) \cdot \underbrace{\cos(2 \cdot 0)}_1 \right] = 0$$

• frontera f_2

$$\lim_{c \rightarrow +\infty} V_\theta(c, \pi/2) = \frac{P}{2} \left[1 + \frac{a^2}{c^2} - \left(1 + 3 \frac{a^2}{c^2} \right) \cdot \overbrace{\cos\left(2 \frac{\pi}{2}\right)}^{-1} \right] = P$$



• frontera f_3

$$V_\theta(a, \theta) = \frac{P}{2} \left[1 + \frac{a^2}{a^2} - \left(1 + 3 \frac{a^4}{a^4} \right) \cdot \cos(2\theta) \right] = P(1 - 2 \cdot \cos(2\theta))$$

$$\frac{\partial V_\theta(a, \theta)}{\partial \theta} = \sin(2\theta) = 0 \begin{cases} \rightarrow \theta = 0 \\ \rightarrow \theta = \pi/2 \end{cases}$$

$$V_\theta(a, \pi/2) = P(1 - 2 \cdot \underbrace{\cos\left(2 \cdot \frac{\pi}{2}\right)}_{\pm 1}) = \begin{cases} -P & (\theta = 0) \\ 3P & (\theta = \pi/2) \end{cases}$$

máximo $\Rightarrow V_{\theta \max} = 3P$ y ocurre en $\theta = \pi/2$ y $r = a$

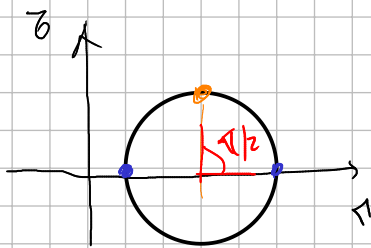
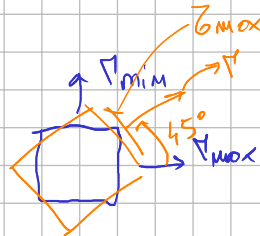
$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Annotations: τ_r points to the square root term, τ_θ points to the same square root term, and τ_{xy} points to the shear stress term.

$$\sigma_{\max} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

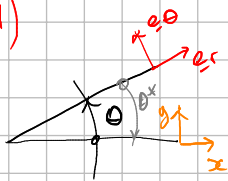
$$\sigma_{\min} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Annotations: τ_r points to the square root term in the $\sigma_{\max}$ equation, τ_θ points to the same square root term, and τ_{xy} points to the shear stress term.



$$2\theta = \pi/2 \Rightarrow \theta = \pi/4$$

(*)1)



$$\{r, \theta\} \rightarrow \{x, y\}$$

$$\theta^* = -\theta$$

$$\underline{T} = \begin{bmatrix} T_x \\ T_y \end{bmatrix} = \begin{bmatrix} \cos(-\theta) & \sin(-\theta) \\ -\sin(-\theta) & \cos(-\theta) \end{bmatrix} \cdot \begin{bmatrix} T_r \\ T_\theta \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} T_r \\ T_\theta \end{bmatrix}$$

$$\underline{T} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} r_r \cos \theta - r_\theta \sin \theta \\ r_\theta \cos \theta - r_r \sin \theta \end{bmatrix} = \begin{bmatrix} r_r \cos^2 \theta - r_\theta \cos \theta \sin \theta - r_\theta \cos \theta \sin \theta + r_r \sin^2 \theta \\ r_r \cos \theta \sin \theta - r_\theta \sin^2 \theta + r_\theta \cos^2 \theta - r_r \cos \theta \sin \theta \end{bmatrix}$$

$$= \begin{bmatrix} r_r \cos^2 \theta + r_\theta \sin^2 \theta - 2r_\theta \sin \theta \cos \theta \\ \frac{(r_r - r_\theta) \sin(2\theta)}{2} + r_\theta \cos(2\theta) \end{bmatrix} = \begin{bmatrix} T_{xx} \\ T_{xy} \end{bmatrix} \rightarrow \text{Coincide con la rotaci3n del tensor de 2do orden de } \{r, \theta\} \rightarrow \{x, y\}$$

$$\underline{T} = \begin{bmatrix} T_{xx} & T_{xy} \\ T_{xy} & T_{yy} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} T_{xx} \\ T_{xy} \end{bmatrix}$$

$$\theta = \lim_{b \rightarrow +\infty} \arctan\left(\frac{y}{b}\right) = 0 \rightarrow \text{Cuando } b \rightarrow +\infty \Rightarrow \theta \rightarrow 0 \wedge 2\theta \rightarrow 0$$

$$\underline{T} = \begin{bmatrix} T_x \\ T_y \end{bmatrix} = \lim_{b \rightarrow +\infty} \begin{bmatrix} \overset{\rightarrow 0}{r_r} \overset{\rightarrow 1}{\cos^2 \theta} + \overset{\rightarrow 0}{r_\theta} \overset{\rightarrow 0}{\sin^2 \theta} - \overset{\rightarrow 0}{2r_\theta} \overset{\rightarrow 0}{\sin \theta \cos \theta} \\ \frac{\overset{\rightarrow 0}{(r_r - r_\theta)} \overset{\rightarrow 0}{\sin(2\theta)}}{2} + \overset{\rightarrow 0}{r_\theta} \overset{\rightarrow 0}{\cos(2\theta)} \end{bmatrix} = \begin{bmatrix} p \\ 0 \end{bmatrix} \rightarrow \text{Verifica CB sobre } f_1$$